

Digital Filtering Explained

User

2.1.2 Digital filtering in details with image

Assistant

2.1.2 Digital Filtering (Detailed Explanation)

English + Nepali Mixup (No Romanized Nepali)

What is Digital Filtering?

Digital Filtering भनेको **digital signal लाई process गरेर unwanted noise, interference, वा unwanted frequency components हटाउने तथा useful information लाई preserve गर्ने process हो।**

यसमा **input signal discrete-time (sampled signal)** को रूपमा हुन्छ र यसलाई **computer, Digital Signal Processor (DSP), microcontroller, FPGA** आदि प्रयोग गरेर process गरिन्छ।

Simple definition:

Digital Filter is a mathematical system that modifies a digital signal by removing unwanted components and keeping desired information.

Illustration of Digital Filtering

Basic Working Principle

Digital filtering मुख्य रूपमा signal processing को technique हो।

Step 1: Input Signal

सबैभन्दा पहिले एउटा signal प्राप्त हुन्छ।

यो signal हुन सक्छ

Voice signal

ECG signal

Temperature data

Image

Audio

Radar signal

Sensor data

तर practical world मा signal प्रायः noise सहित आउँछ।

Example

Original Voice + Background Noise

Step 2: Sampling

Real-world signal continuous हुन्छ।

Computer continuous signal बुझ्दैन।

त्यसैले

Continuous Signal

↓

Sampling

↓

Discrete Signal

बन्छ।

यसलाई Analog-to-Digital Converter (ADC) ले convert गर्छ।

Step 3: Digital Filter

अब digital signal filter भित्र जान्छ।

Filter ले mathematical calculations गर्छ।

यसले

✓ unwanted frequencies हटाउँछ

✓ desired frequencies राख्छ

✓ signal smooth बनाउँछ

✓ distortion कम गर्छ

Step 4: Output Signal

Output मा cleaner signal प्राप्त हुन्छ।

Noisy Signal

↓↓↓

Digital Filter

↓↓↓

Clean Signal

Real Life Example

Imagine तपाईं mobile बाट आफ्नो साथीसँग call गर्दै हुनुहुन्छ।

Background मा

fan

vehicle

people talking

wind noise

सबै आवाज आइरहेको छ।

Digital filter ले

✗ unwanted background sound हटाउँछ।

✓ तपाईंको voice preserve गर्छ।

त्यसैले अर्को व्यक्तिले clear आवाज सुन्छ।

Another Example (Music)

Suppose एउटा song play भइरहेको छ।

Recording गर्दा

Song

+

Static Noise

+

Electrical Noise

record भयो।

Digital filter ले

Noise remove गर्छ।

Output

Clean Music

Another Example (Image)

Suppose एउटा image मा random dots छन्।

Before



☺.....
.....☺...
.....

After filtering



☺

Noise कम हुन्छ।

Image clearer हुन्छ।

Why Digital Filtering is Needed?

Digital filtering किन आवश्यक छ?

1. Noise Removal

सबैभन्दा common application।

Sensor data मा

electrical noise

thermal noise

random noise

remove गर्न प्रयोग हुन्छ।

2. Signal Enhancement

Weak signal लाई clear बनाउँछ।

Example

Medical ECG signal

3. Data Compression Preparation

Noise हटाएपछि compression राम्रो हुन्छ।

4. Image Processing

Blur हटाउन

Sharpen गर्न

Sharpening

Smooth गर्न

5. Audio Processing

Echo reduction

Noise reduction

Speech enhancement

6. Communication Systems

Receiver मा

Received signal

↓

Filtering

↓

Original message

recover हुन्छ।

Characteristics of Digital Filters

Digital filter का मुख्य characteristics

High Accuracy

Mathematical computation भएकाले accuracy धेरै हुन्छ।

Programmable

Software change गरेर filter change गर्न सकिन्छ।

Hardware change गर्नुपर्दैन।

Stable

Properly designed filter धेरै stable हुन्छ।

Repeatable

एकै input मा सधैं एउटै output दिन्छ।

Flexible

Different applications अनुसार modify गर्न सकिन्छ।

Easy to Design

Software बाट design गर्न सकिन्छ।

MATLAB

Python

DSP tools

Components of Digital Filtering System

Input Signal

↓

ADC

↓

Digital Filter

↓

DAC (optional)

↓

Output

Input Signal

Real-world signal

ADC

Analog लाई Digital मा convert गर्छ।

Digital Filter

Signal process गर्छ।

DAC

यदि analog output चाहिएको छ भने प्रयोग हुन्छ।

Output

Filtered signal

Mathematical Representation

Digital filter लाई सामान्यतया यसरी लेखिन्छ

$$y[n] = \text{Filter}(x[n])$$

$$y[n] = \text{Filter}(x[n])$$

Where

$x[n]$ = Input Signal

$y[n]$ = Output Signal

Example Output Signal

Example

Input

10 12 100 11 13

यहाँ

100 एउटा noise हो।

Filtering पछि

10 12 12 11 13

Noise हट्यो।

Types of Digital Filters

Digital filter दुई प्रमुख प्रकारका हुन्छन्।

1. FIR (Finite Impulse Response)

Features

Always stable

Simple design

No feedback

More calculations

Linear phase possible

Block

Input

↓

FIR Filter

↓

Output

2. IIR (Infinite Impulse Response)

Features

Feedback पाउँछ

Feedback loop

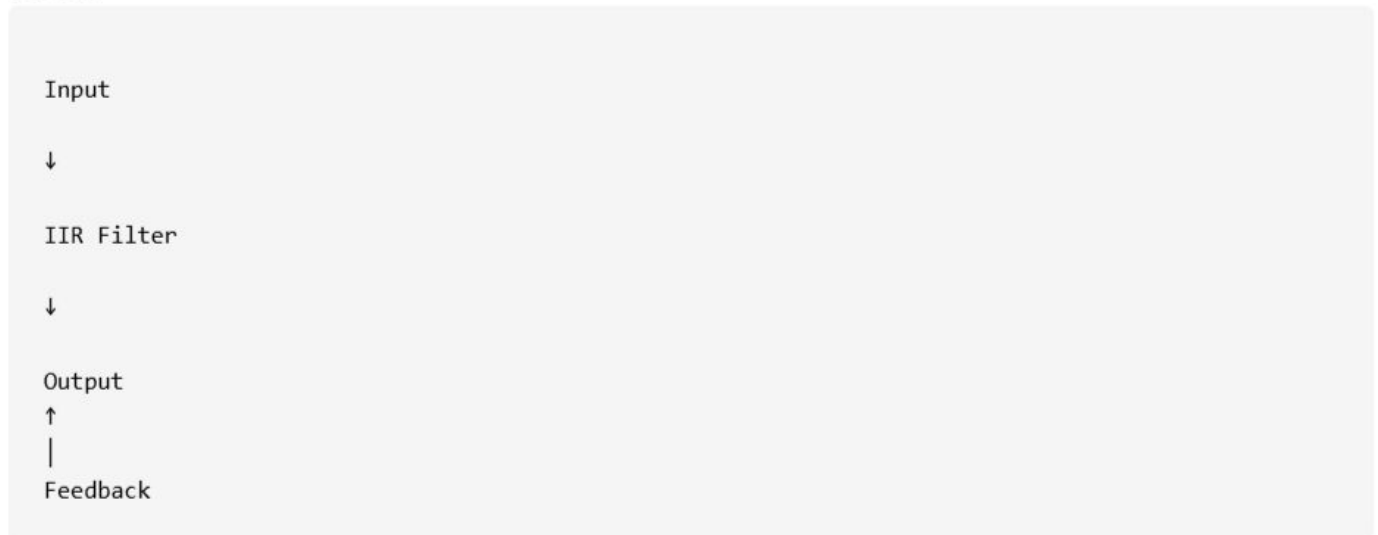
Faster

Less computation

High efficiency

Stability carefully maintain गर्नुपर्छ

Block



Frequency Based Types of Filters

Digital filter frequency अनुसार पनि classify गरिन्छ।

1. Low Pass Filter (LPF)

Low frequency पास गर्छ।

High frequency हटाउँछ।

Applications

Audio smoothing

ECG

2. High Pass Filter (HPF)

High frequency पास गर्छ।

Low frequency हटाउँछ।

Applications

Edge detection

Image processing

3. Band Pass Filter (BPF)

Specific frequency range मात्र pass गर्छ।

बाकी remove गर्छ ।

Applications

Radio

Wireless communication

4. Band Stop Filter (Notch Filter)

Specific frequency हटाउँछ ।

बाकी सबै pass गर्छ ।

Applications

50 Hz power line noise removal

60 Hz interference removal

Applications of Digital Filtering

Digital filtering धेरै क्षेत्रमा प्रयोग हुन्छ ।

Field	Application
Audio Processing	Noise reduction, echo cancellation
Image Processing	Blur removal, sharpening, smoothing
Medical	ECG, EEG, MRI signal processing
Mobile Communication	Voice enhancement
Satellite Communication	Noise removal
Radar Systems	Target detection
Sonar	Underwater signal processing
Robotics	Sensor filtering
IoT	Sensor data cleaning
Video Processing	Frame enhancement
Speech Recognition	Voice cleaning
Artificial Intelligence	Data preprocessing

Advantages of Digital Filtering

High accuracy

Programmable

Stable

Stable

Flexible

Repeatable results

Easy implementation

Low maintenance

Noise reduction

Signal enhancement

Easy integration with computers

Disadvantages of Digital Filtering

ADC र DAC आवश्यक पर्न सक्छ।

High-speed processing का लागि powerful processor चाहिन सक्छ।

Sampling error हुन सक्छ।

Processing delay (latency) आउन सक्छ।

High-order filters मा computation बढी हुन्छ।

Real-Life Applications

Mobile Phone

Voice noise reduction।

WhatsApp Call

Background noise हटाउँछ।

Bluetooth Earbuds

Environmental noise cancel गर्छ।

Hospital ECG Machine

Patient को ECG signal सफा बनाउँछ।

CCTV Camera

Image clearer बनाउँछ।

YouTube Audio

Audio enhancement।

Smart Speaker

Voice recognition अघि noise हटाउँछ।

Self-Driving Cars

Camera र LiDAR sensor data filtering।

Summary

Digital Filtering भनेको **digital signal लाई mathematical operations प्रयोग गरेर process गर्ने प्रविधि** हो, जसले **unwanted noise हटाएर desired signal लाई सुरक्षित राख्छ**। यो **audio processing, image processing, communication systems, medical instruments, radar, robotics, IoT, speech recognition** लगायत धेरै क्षेत्रमा प्रयोग हुन्छ। प्रमुख प्रकार **FIR र IIR** हुन्, जबकि frequency response अनुसार **Low Pass, High Pass, Band Pass, र Band Stop (Notch) filters** प्रयोग गरिन्छ। Proper digital filtering ले signal quality, accuracy, र reliability उल्लेखनीय रूपमा सुधार गर्छ।
